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SHORT COMMENTARY

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CHANGE IN SEA ICE TRENDS IN THE CANADIAN INTERNATIONAL POLAR YEAR PROGRAM

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Abstract :

The comprehensive synthesis of scientific sea ice research and traditional findings from the Canadian IPY project from 2007 to 2009. In particular, an overview of basin-wide and regional sea ice trends is presented along with collected Inuit knowledge about sea ice. in communities in northern Canada. We focus on recently observed changes in sea ice and discuss some of the causes of this change, including atmospheric and ocean forces, dynamic forces on the ice, and thermodynamics.

Keywords : Scientific sea ice, Atmospheric, Dynamic force

Commentary:

Related findings include: 1) In Amundsen Bay, at the western end of the Northwest Passage, open water persists longer than usual, and winter sea ice is thin and mobile. 2) Large areas of summer sea ice degrade significantly in summer and Cyclones play an important role in fault zones, that is, open water areas between ice floes and solid ice on land.

They slow the formation of new ice and the growth of permanent ice. 4) Feedbacks related to increased open ocean duration, long-period wave occurrence, increased open ocean roughness, and snowfall precipitation may account for some of the observed decline in perennial sea ice. There is. 5) It has been observed that during winter the atmosphere is generally stable, preventing vertical entrainment of moisture to the earth's surface [1].

Scientists work to understand the myriad manifestations of landscape by studying the physical processes that produce its structures and the ways in which people live in and manipulate landscapes (e.g., landscape architecture). have spent thousands of years on. In the Arctic, Inuit have lived in and on the icy surface for centuries, using it for travel and cultural exchange.

Traditional knowledge systems represent knowledge gained through the experiences and observations of different individuals in their lives. Each body of knowledge (i.e., traditional knowledge and Western science) has studied ice landscapes in depth. In what follows, we use the term "icescapes" and specifically examine the interface between Western science and Inuvialuit knowledge in order to understand how these "icescapes" function [2].

The purpose of this paper is to cover important parts of sea ice research, and to present findings from several Canadian-led IPY projects, with a particular focus on the 'causes' of observed sea ice change and variability. The purpose is to provide an overview and summary.

IPY's largest sea ice research project was the Circumpolar Fault Lead System Study (IPY-CFL), which focused on large-scale fault leads in the Southern Ocean and Amundsen Bay. The IPY-CFL study is the first time that a wintering research icebreaker has continued to move within the lead polynya, providing the opportunity to study ice development in the region throughout the annual sea ice cycle.

Also presented will be the results of the IPY project "Changes and Changes in the Canadian Cryosphere," which classified changes in sea ice and snow cover in the Canadian Islands, and the long-term study THORPEX Arctic Weather and Environmental Prediction Initiative [3]. A program led by the World Meteorological Organization that aims to improve the accuracy of weather and environmental forecasts. Through Canada's IPY program, both Western science and traditional knowledge have been recognized as complementary knowledge systems that can understand the mechanisms and consequences of these changes. Below we summarize new discoveries about the causes of sea ice change and variability. This data, collected from both Western science and traditional knowledge through interviews, was a direct result of Canada's participation in the International Polar Year (IPY).

Community members have also reported changes in the nature of the pressure ridges around the community. The pressure bumps are less and not as thick or high as before. Rising temperatures appear to be affecting ice formation, which is no longer exposed to the conditions necessary to thicken as it once did, resulting in increased talus formation in some areas, Multi-year ice formation is blocked. Ridges are made up of smaller ice blocks that break down and pile up more easily, so they melt faster. Strong winds break up the thin ice and create tall, impassable piles of debris. The formation of ridges and the magnitude of ice accumulation also depend on the strength of winds and currents. At tidal changes are also thought to influence ice formation on pressure ridges [4].

Three communities are located near areas that are typically known to have open water, rifts, or pressure ridges. Open water can be found not only in deep waters outside the bay, but also in shallow waters characterized by strong currents. The possibility of more ice mountains in flatter areas was also discussed in. This complements Sachshaber's observations showing increased ice movement in shallow and open areas. Residents in each of the three communities said the impact of ice debris on travel was significant. Local residents had to change their travel routes or stop traveling on the ice altogether because the ice was no longer smooth to drive on. Even in late winter when the ice is solid (e.g. from January to March), being on the ice is extremely dangerous for travelers. Ice fragments are also known to harm animals. Areas with ice patches can make it more difficult for polar bears to access seals. Even when hunting, bears have difficulty penetrating the ice.

Similar observations have been reported in other of his IPY projects involving members of northern communities in research based on traditional knowledge. The project, entitled the Inuit Sea Ice and Occupancy Project (ISIUOP), will survey communities in Nunavut and Nunavik in the central and eastern Canadian Arctic, drawing on local expertise and experience with sea ice. have been documented and compiled into a map [1]. This study characterized seasonal sea ice conditions, the extent and area of sea ice use, the nature and location of significant sea ice hazards, important fishing grounds, and traditional and current ice (and land) routes. I did. Observations showed a similar trend. Sea ice freezes in late fall, sea ice break-up occurs early in spring, and sea ice is generally less reliable and predictable. They also found that the extent to which changes affected sea ice activity varied from community to community and, in some cases, from hunter to hunter [5].

As a result, major changes have occurred in the design of sea ice corridors and sampling sites used by communities.

As part of IPY, analyzed weekly ice maps from the Canadian Glacier Authority Digital Archive (CISDA) for the period 1960 to 2008 to determine trends in summer sea ice in the Canadian Arctic. We looked at variability. They concluded that summer sea ice extent in Hudson Bay decreased most between 1960 and 2008, decreasing by about 11% per decade [2].

They also found that sea ice extent decreased by about 2.9% per decade in Canada's Arctic Islands, by 8.9% per decade in Baffin Bay, and by 5.2% per decade in the Beaufort Sea. Their findings provided evidence that rising Arctic surface temperatures may further reduce summer sea ice extent, thereby increasing maritime transport through the Canadian Arctic in the near future.

The annual trend in Baffin Bay from 1979 to 2006 was around -8000 ± 2000 km2 per year, with a significant decrease in ice concentration after 1996. In contrast to the past five years when the Nares Strait was relatively ice-free, Baffin Bay became ice-free in mid-August 2010 and an ice bridge formed across the Nares Strait in mid-February 2011 [6].

A slight increase in ice concentration in the eastern Beaufort Sea and a significant decrease in ice concentration in the western Beaufort Sea between the summers of 1979 and 2006 were also observed in studies of the relationship between ice and atmospheric phenomena [3]. Thinning of Arctic sea ice, particularly multi-year ice loss in the Beaufort Sea, has been documented in other studies.

Increasingly intense hurricanes and vast expanses of water represent a linked response to climate change. An increase in the area of wave-generating waves is associated with a decrease in the extent of Arctic sea ice in summer and may lead to an increase in the propagation of wave energy within the ice pack.

In Antarctica, large wave intrusions have been observed, causing sea ice floes to break up into smaller ice floes. At the year ice (MYI) site, approximately 250 km from the ice edge, document observations of the initial propagation of cyclone-driven longwave waves into the Arctic MYI (ice floes). These swells were generated by sustained intense cyclonic activity in the Beaufort Sea from September 1 to 6, 2009, which caused widespread destruction of the MYI packing almost instantaneously. The ice floes west of this event were severely disintegrated and rotten ice. This may be the key to why these ridges spread so far into the ice floes. A cyclone that destroys large areas of the open ocean in the Arctic Basin can lead to the occurrence of larger-than-normal storm surges, such as those observed during IPY [7].

Even the thickest MYI ice floes, are weakened by warm summer temperatures (near freezing) and can break through storm waves. Breaking the ice reduces albedo (more open water), increases the surface area of ice floes exposed to the ocean by a factor and traps radiation, accelerating melting [2]. New sea ice formation in the fall can weaken the ice surface and make it more susceptible to continued fracturing and trenching. Stronger ripples could act as a positive feedback (promoting ice thickness), while frequent formation and refreezing of sea ice could cause further heat and moisture influx into the atmosphere. there is.

The vertical structure of the boundary layer is also a factor in the atmospheric forcing of sea ice. The IPY-CFL project operates the radiometric microwave profiling radiometer (MWRP) MP-3000A© aboard the research icebreaker CCGS Amundsen, which measures atmospheric boundary layer (ABL) temperature (°K) and absolute humidity (x) Profiles have been collected. 10-3 kg m−3) over Amundsen Bay in the Arctic Circle of western Canada during the winter and summer of 2008. Comparison of these profiles with contemporaneous radiosonde data reveals that these profiles are reliable from the surface to depths of 1500-2000 m. used a time series of hourly MWRP soundings to describe the general characteristics of the ABL [8]. Both the winter mean (JFM) and spring mean ABL potential temperature profiles(AMJ) exhibit a shallow (350 m below the surface) thermal internal boundary layer (IBL) with a rising inversion (350-1000 m in winter; 350 to 800 m). m) was crowned in the spring). The presence of IBL was attributed solely to convective turbulence rather than shear-induced mixing.

Conclusion:

The stable top of the ABL is thought to weaken the turbulent exchange of sensible and latent heat between the free atmosphere and the surface. This thermal profile can also result in large vertical wind shear while minimizing downward momentum transfer between free-air synaptic winds and sea ice. During the summer, a stable IBL facilitated the downward flow of sensible and latent heat to the sea ice surface, while the rising mixed layer, probably in response to free air advection, added heat and momentum to the sea ice topography.

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